# WASP-52b

**R-0318**

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-52_221129` · created 2026-07-18T01:22:11+00:00

**BENCHMARK: NON-DETECTION**

## Results

| | |
|---|---|
| Mid-Transit Time (Tmid) | 2459912.679 +/- 0.0072 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*) | 0.205 +/- 0.063 |
| Transit depth (Rp/Rs)^2 | 4.22 +/- 2.59 [%] |
| Orbital Inclination (inc) | 83.65 +/- 2.78 |
| Airmass coefficient 1 (a1) | 1.56 +/- 0.87 |
| Airmass coefficient 2 (a2) | -0.33 +/- 0.35 |
| Scatter in the residuals of the lightcurve fit is | 54.87 % |
| Best Comparison Star | #3 - [429, 362] |
| Optimal Aperture | 2.48 |
| Optimal Annulus | 8.12 |
| Transit Duration (day) | 0.057 +/- 0.03 |

## Accuracy vs published ephemeris (ExoClock/NEA)

| | |
|---|---|
| Rp/R■ fit vs published | 0.205 ± 0.063 vs 0.1646 (deviation 0.55σ) |
| Duration fit vs published | 0.057 d vs 0.0758 d (deviation 0.54σ) |
| Mid-time O–C | 18.1 min |
| Depth SNR | 2.8 |
| Residual scatter | 54.87 % |

## Light curve

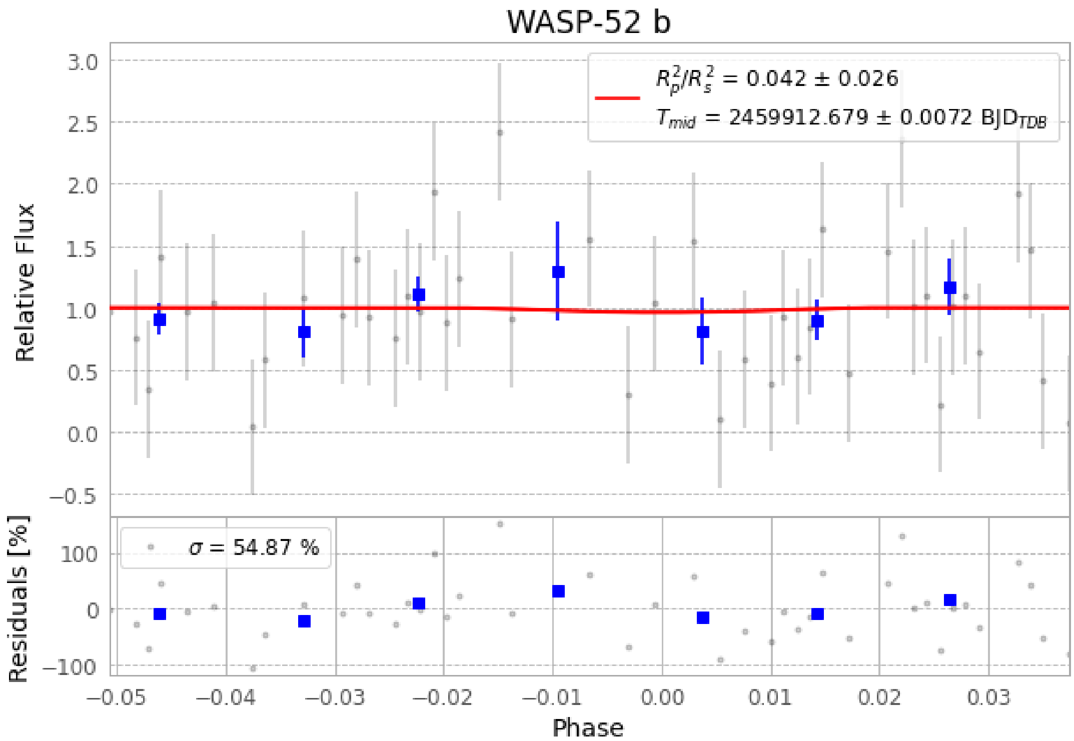


Figure 1 — FinalLightCurve_WASP-52 b_2022-11-28.png (SHA-256 6e7d7cb6acd97232…)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [327, 105], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 ac9360e9522386ee…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1bbef5c`

## Data provenance

76 FITS frames (50 MB), `WASP-52221129020615.FITS` → `WASP-52221129055115.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-221129.log (91a5d7157bee…), FinalLightCurve_WASP-52 b_2022-11-28.png (6e7d7cb6acd9…), FinalLightCurve_WASP-52 b_2022-11-28.csv (ce2d08ae9169…)

## Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)